

CPIS-352 Syllabus

Catalog Description

CPIS-352 IS Applications Design and Development

Credit: 3 (Theory: 3, Lab: 2, Practical: 0)

Prerequisite: CPIS-351

Classification: Department Required

The objective of this course is to explore the design, selection, implementation, and management of enterprise IT solutions. The focus is on applications and infrastructure and their fit with the business. Topics include frameworks and strategies for infrastructure management, system administration, data/information architecture, content management, distributed computing, middleware, legacy system integration, system consolidation, software selection, total cost of ownership calculation, IT investment analysis, and emerging technologies. These topics are addressed both within and beyond the organization, with attention paid to managing risk and security within audit and compliance standards. Students also hone their ability to communicate technology architecture strategies concisely to a general business audience.

Class Schedule

Lab/Tutorial 90 minutes 1 times/week

Meet 50 minutes 3 times/week or 80 minutes 2 times/week

Textbook

Marc Lankhorst, , "Enterprise Architecture at Work", Springer; 4 edition (2017-02-12)

ISBN-13 9783662539323 ISBN-10 3662539322

Grade Distribution

Week	Assessment	Grade %
4	Quiz 1	5
5	Graded Lab Work 1	2.5
6	Graded Lab Work 2	2.5
8	Graded Lab Work 3	2.5
9	Exam 1	30
12	Graded Lab Work 4	2.5
12	Quiz 2	5
14	Group Project	20
16	Exam	30

Last Articulated

February 5, 2021

Relationship to Student Outcomes

1	2	3	4	5	6
x	x	x		x	x

Course Learning Outcomes (CLO)

By completion of the course the students should be able to

1. Utilize the core concepts and key terminologies of TOGAF (1)
2. **Compare contemporary Enterprise Architecture frameworks like TOGAF, ZIFA, E2AF, DoDAF and EAP (2)**
3. **Articulate the foundational elements of Enterprise Architectures i.e. Enterprise Continuum, Building Blocks, Artifacts, Deliverables (2)**
4. Utilize Enterprise Architecture concepts and practices (1)
5. Articulate how enterprise architecture supports the strategic objectives of businesses (3)
6. Demonstrate the use of Architecture Development Method cycle and the objectives of each phase, and how to adapt and scope the ADM (6)
7. **Apply the core concepts of data/information architecture and evaluate existing data/information architecture designs (2)**
8. Observe the use of Enterprise Continuum; its purpose, and its constituent parts (3)
9. **Describe and discuss the benefits and risks of service oriented architecture (2)**
10. **Identify views, viewpoints, stakeholders etc in Information systems and ADM and their role in communication (2)**
11. Represent how each phase of the ADM contributes to the success of enterprise architecture (5)
12. Describe a variety of frameworks for enterprise architecture analysis and decision making (6)
13. **Use architectural modelling tools and relate models with real business problems (2)**
14. Demonstrate how Architecture Governance contributes to the Architecture Development Method life cycle (6)
15. Determine the integration of emerging technologies for the architecting trends (6)
16. Define the role of audit and compliance in enterprise architecture (6)

Coordinator(s)

Dr. Aisha Alsobhi, Assistant Professor

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Topics Coverage Durations

Topics	Weeks
Explain the difference between Internet and Web and Introducing the ASP.NET development environment	1
Introduction to HTML5	2
Introduction to CSS3	2
Introduction to JavaScript	1
Adding Dynamic Visualization to Web site by using JavaScript	1
Client side Form validation and dynamic form by using JavaScript	1
Server side Form Handling in ASP.NET	1
Interacting with Database and Web sites in ASP.NET	2
Search and Advance Search in ASP.NET	1
Using Advance Controls Such as Gridview	1
Using Cookies in ASP.NET	1
Using Sessions in ASP.NET	1